

SmartHearing: A Multimodal Audio Large Language Model Based Personalized Intelligent Hearing Assistance System

Abstract

Traditional hearing aids mainly focus on speech enhancement or overall noise reduction, although sound importance changes with the acoustic scene. To address this problem, this research proposes SmartHearing, which combines LALM-based scene understanding with audio separation and processing.

SmartHearing contains two paths. Its perception path uses Qwen3.5-Omni to analyze sound objects, while CLAP and a Query-space Adapter encode the target descriptions. Its execution path uses LASS to separate sounds and an Audio Processor to adjust their gains. The system also adaptively selects the chunk size to balance audio quality and latency.

The three main innovations are: **1)** a low-latency LALM-based perception–execution dual-path architecture for dynamic and complex acoustic scenes; **2)** an LALM-to-LASS Adapter that corrects the representation shift between LALM-generated sound descriptions and the original LASS training queries; and **3)** a sound-adaptive time-window mechanism that selects a 4–10-second window for the next processing cycle based on sound category, recognition performance, danger level, and processing time.

Adapter V2 was trained on 3,500 mixed-audio samples and validated on 700 samples. In the external baseline experiment, compared with Base LASS, developed by a research team including Alibaba Speech Lab, it improved SNRi and SI-SDRi by 0.957 dB and 1.015 dB, respectively, while the separate historical ablation also showed an improvement. These comparisons use different input and correction-strength settings. To examine generalizability, another Adapter was trained and validated without any ESC-50 target audio and evaluated on newly constructed ESC-50 mixtures. On 145 unseen samples, it still outperformed Base LASS by 1.323 dB in SNRi and 1.192 dB in SI-SDRi, showing that the improvement in the main separation metrics can transfer to an unseen dataset source.

To balance audio quality and processing time, a chunk-size experiment evaluated 600 mixtures covering six representative safety-related sounds, ten interference types, and five input SNR conditions. Each mixture was tested under ten chunk sizes, forming 6,000 experimental units. The results show that there is no single optimal window for all sounds: when recognition performance and processing time are considered together, the six sound categories select windows from 4 to 10 seconds. This difference indicates that sound duration and periodicity influence the required temporal information. These window choices were selected and evaluated on the same mixtures and still require independent validation.

Finally, a listening questionnaire provided 101 complete response records. SmartHearing was selected as the condition in which the target sound was easiest to notice in 50.5% of the choices, compared with 28.1% for Base LASS and 21.5% for the unprocessed mixture. This is preliminary response-record evidence of target salience; participant recruitment and listening compliance remain to be independently verified.

Keywords: personalized hearing assistance; large audio language model; language-queried audio source separation; Query-space Adapter; environmental sound enhancement; adaptive chunk size

Contents

Abstract	I
1 Introduction	1
2 Related Work	3
2.1 Hearing Assistance and Selective Sound Enhancement	3
2.2 Acoustic Scene and Event Analysis	3
2.3 Language-Queried Audio Source Separation	4
3 Methodology	4
3.1 Problem Definition	4
3.2 Overall Architecture	5
3.3 LALM-Based Perception Path	6
3.4 Perception–Execution Interface	6
3.5 Execution Path	7
4 Experiments and Results	10
4.1 Experimental Setup	10
4.2 Baselines and Evaluation Metrics	11
4.3 Main Experimental Results	12
4.4 Ablation and Further Analyses	16
4.5 Case Study	22
5 Conclusion	25
6 Acknowledgements	27
References	29
Appendix	31

1 Introduction

With the increasing of aged and hearing-impaired population, audio assistive devices have taken an important role in the field of medical restoration and intelligent health caring. In reality, hearing impaired population often encounter diverse and complex audio environment such as restaurant, shopping mall, multi-speaker meetings, etc., in which background noise, mixed sound, overlapping speaking often exist. Conventional hearing aids primarily amplify human voices, other sounds that carry important information may be omitted, such as car honking on streets, causing fatal danger. Therefore, conducting research to explore a method of extracting and enhancing specific sound pattern in complex environments has a clear real-life necessity and application value. This method may also be used in auditory attention system in embodied AI and intellectual meeting, in-vehicle agents, providing an innovative and crucial method for audio processing.

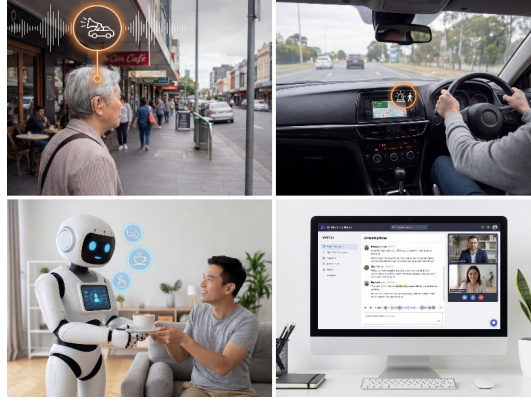


Figure 1.1: Real-World Application of proposed model, showing four scenarios: hearing aids, embodied ai interaction, intellectual in-vehicle and meeting agent.

Over the past few decades, audio signal processing has transformed from classical statistical modelling to a deep-learning-driven end-to-end paradigm. Early environmental audio analysis was mainly built upon predefined Acoustic Scene Classification, or ASC, i.e., labelling an audio recording with a specific label, such as “office”, to represent its surrounding environment. However, real-world situations often have significant complexity, flexibility, and scenario-specificity. Take a home environment as an example: multiple acoustic sources usually exist at the same time, such as an infant crying, keyboard tapping, human speech, and other background noises. In such a dynamic and complex environment, relying only on acoustic scene classification may omit many crucial sound characteristics. Also, ASC can mainly identify scenes within its training data, which limits its generalizability and adaptability to new scenarios. Thus, audio signal processing should not only classify a specific environment, but also identify relevant acoustic factors and sound patterns that can indicate the characteristics of

the environment[3, 20]. Moreover, discovering the relationships between different audio components and making predictions about possible future events based on current audio input are also important, as they may further reveal the underlying interactions in the scene and help the system respond to potential danger.

Auditory Scene Analysis, or ASA, attempts to model the human auditory attention mechanism, which does not treat all external signals equally, but instead prioritizes them according to environmental context, behavioral objectives, and potential danger [1, 2, 19]. For example, on a crowded street, pedestrians can usually ignore the continuous background noise caused by traffic flow, yet respond immediately to an abrupt braking sound or warning horn. This importance-driven mode of processing audio input requires a model not only to recognize a particular sound signal, but also to analyze its contextual significance and anticipate what may occur next [9].

Recent developments in Large Audio Language Models (LALMs) have made it possible to integrate high-level semantic reasoning with audio signal enhancement.[6, 7, 15, 22]. LALMs such as Audio Flamingo 2 and 3, Qwen2-Audio, and Qwen3-Omni have demonstrated strong performance in audio understanding and complex causal-relationship reasoning[6, 7, 15, 22]. These models provide a basis for developing a system that uses the reasoning capabilities of LALMs to identify salient acoustic patterns in complex environments and guide the following audio enhancement process.

This project intends to address several core limitations of conventional hearing aids and scene-classification models: (1) difficulty in processing complex acoustic environments containing mixed sounds, (2) lack of reasoning about the contextual significance of different sounds during sound extraction and enhancement, (3) limited generalizability to unprecedented scenarios, and (4) limited ability to predict potential dangers from current audio information.

Therefore, this project aims to develop a personalized sound extraction and enhancement system that can adapt to complex environments. It uses LALMs for acoustic-component recognition and contextual reasoning, and then integrates their outputs with an audio enhancement model to produce a comprehensive intelligent hearing-assistance system. The use of LALMs allows the system to conduct (1) audio-component detection, (2) reasoning about the contextual importance of different sounds, (3) adaptation to new acoustic scenarios, and (4) prediction of potential future events based on current audio input. Since the system is designed according to real-life needs, its robustness and real-time performance are also important parts of the research.

2 Related Work

2.1 Hearing Assistance and Selective Sound Enhancement

Modern hearing assistance systems usually combine frequency-dependent amplification, dynamic-range compression, beamforming, directional processing, and noise reduction, in order to improve users' access to important acoustic information [10, 14]. Among these objectives, speech intelligibility in noisy environments has been an important target for both research and practical systems [10, 14].

2.2 Acoustic Scene and Event Analysis

Acoustic Scene Classification (ASC) models categorize audio environments into predefined scenes, predominantly using either lightweight CNNs or Transformer-based models [3]. CNN-based models transform acoustic signals into time-frequency representations and discover their local structures, such as short-term repetitive patterns. TF-SepNet models the time and frequency dimensions separately, allowing the model to capture their different characteristics [23]. AST is an attention-based ASC model that uses a Transformer to study the relationships between audio tokens [8]. However, ASC methods usually attach only one scene label to an audio recording. In a complex real-life scene, multiple sound signals may exist at the same time. This fixed categorization makes ASC less flexible when it encounters complex or unprecedented environments.

Auditory Scene Analysis (ASA) is the ability to organize a complex auditory mixture into meaningful events, which is fundamental for auditory perception [1]. The early computational model proposed by Brown and Cooke used hand-crafted acoustic rules and grouping principles to separate mixed sound sources [2]. Later, Shamma et al. introduced temporal coherence, stating that acoustic features that co-modulate over time are likely to originate from the same sound source [19]. However, existing ASA models often focus on individual functions, such as separation or event detection, rather than forming a complete framework. Moreover, multiple overlapping sound sources in real-world environments make contextual analysis more difficult, which also limits their ability to support the prediction of potential dangers.

Sound Enhancement and Target Separation models attempt to recover a desired sound component from noisy or mixed audio. DeepFilterNet2 is a lightweight model designed for real-time full-band speech enhancement on embedded devices [17]. Personalized speech enhancement methods introduce information about the target speaker. For example, VoiceFilter uses a prerecorded voiceprint embedding to guide the model to extract a target voice from a mixed

acoustic environment [21]. Conv-TasNet directly separates overlapping speech signals in the time domain through a temporal convolutional network [12]. Despite their advantages, most of these methods focus on speech, predefined targets, or fixed speaker identities. Therefore, their target definition is less flexible when the sound that should be enhanced changes according to the environmental context.

2.3 Language-Queried Audio Source Separation

Language-queried audio source separation allows a model to extract a target sound according to its natural-language description [11]. For example, CLAP maps text and audio into a shared embedding space, allowing a natural-language query to be used as the semantic condition of a downstream separation model [4].

Compared with fixed-target separation, this mechanism has higher flexibility. However, it also introduces a new problem that the same sound object can be described in many different natural-language forms. For example, the same dog barking sound may be described as “dog barking,” “barking dog,” or “dog sound.” These descriptions may produce different text representations even though they refer to the same acoustic target [4].

On the other hand, the distribution of queries generated or normalized by an LALM may not be completely consistent with the distribution of language queries used during the original training of the LASS model. Therefore, the same target sound may be represented differently when the query comes from a user, a dataset label, or an LALM-generated output.

3 Methodology

3.1 Problem Definition

In real acoustic scenes, multiple sound sources usually exist at the same time, and the input audio of the processing window at time t can be denoted as x_t . Traditional hearing aids usually focus on overall enhancement, speech enhancement, or general noise reduction. However, in real scenes, the importance of different sounds depends on their acoustic context and the sound sources themselves. Therefore, this research needs to address the following problem: how to conduct personalized audio processing under complex and dynamically changing acoustic scenes?

The output audio should increase the salience of safety-related sounds, while preserving necessary speech communication and environmental information and reducing irrelevant interference and background noise. Since a hearing assistance system needs to respond to suddenly

occurring dangerous sounds in time, this problem is also constrained by both audio quality and processing latency. SmartHearing uses an LALM-based perception path to understand the current scene, uses the Perception–Execution Interface to convert semantic requirements into executable text embeddings, and then uses the LASS-based execution path to separate and process different sounds. Finally, it outputs the processed audio. Dual-path asynchronous execution is used to minimize processing latency.

3.2 Overall Architecture

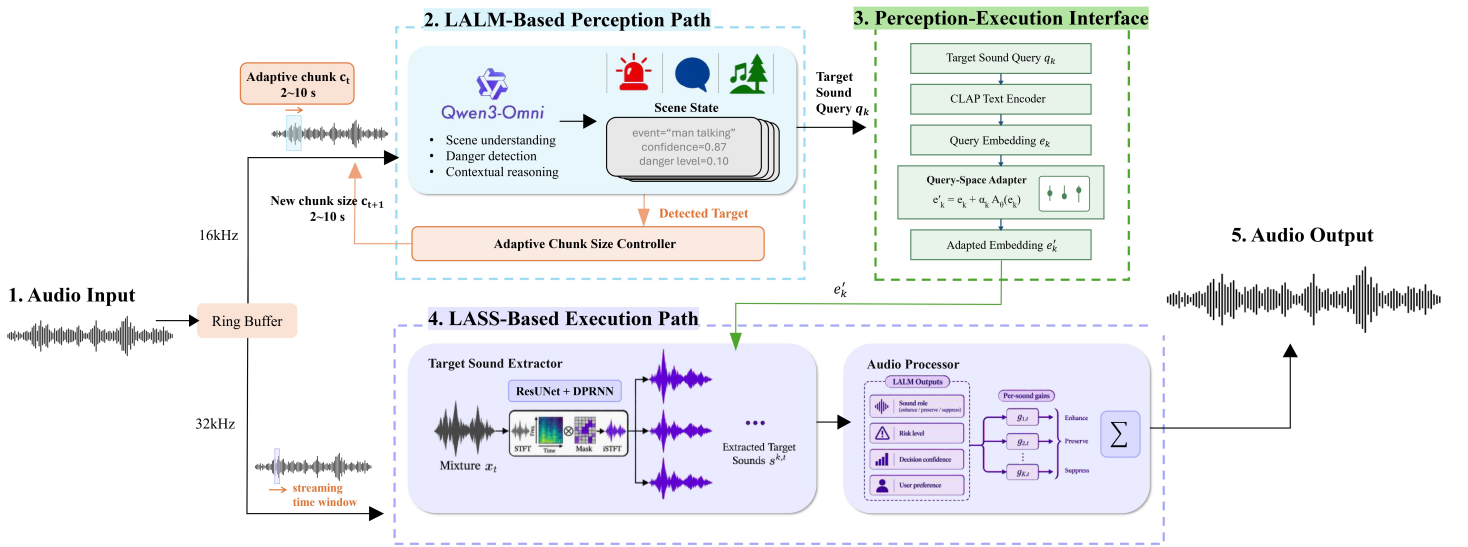


Figure 3.1: Overall architecture of the proposed SmartHearing system.

The overall architecture of SmartHearing is shown in Figure 3.1. The SmartHearing hearing assistance system combines the audio understanding ability of large models to conduct an in-depth analysis of acoustic scenes, identify the functions of sound sources, and analyze the type of the acoustic scene. Based on the acoustic scene, it then judges the importance of each sound source to determine how different sound sources should be processed. The overall system consists of a perception path, an execution path, and an interface connecting them. The input mixed audio is denoted as $x(t)$. The audio first passes through a ring buffer and is separately re-sampled to 16 kHz and 32 kHz. They are then respectively sent to the LALM-based perception path for analysis and to the execution path for sound extraction and processing.

The perception path first uses the LALM to produce a structured acoustic-scene analysis, including the role of each sound object in the current scene, its danger level, and the confidence of the decision. After obtaining the structured analysis, the LALM further deepens its reasoning and summarizes the sounds that should be enhanced or suppressed in the scene.

The execution path first converts the knowledge-based language descriptions produced by the LALM into text embeddings. The embeddings are then corrected by the trained Adapter. Finally, a neural network integrating ResUNet and DPRNN is used to extract specific sounds and perform audio enhancement or suppression. Multiple processed sound sources are then sent to the mixer together with the original background sound. According to the role, risk, and confidence of each sound, this module adaptively adjusts the salience of each sound and outputs the audio. To achieve real-time streaming processing, the LALM and the execution layer separately conduct model inference asynchronously on a longer rolling context, so that model inference will not block audio processing.

3.3 LALM-Based Perception Path

The perception path uses Qwen3.5-Omni to analyze windows of 16 kHz audio. Unlike traditional Acoustic Scene Classification, which only outputs a single scene label, the LALM in the system decomposes the scene into multiple sound objects and analyzes the role, confidence, and danger level of each sound object in the current scene for the following sound-role analysis.

Through iterative prompt engineering and prompt experiments, the output of the LALM is adjusted into a systematic and structured scene-state output. Each sound object is represented as

$$o_k = (\text{sound}_k, \text{role}_k, d_k, p_k, u_k), \quad (3-1)$$

where sound_k is the name of the sound, role_k is the processing role of the sound in the current scene, d_k is its danger level, p_k is the recognition confidence, u_k is its relevance to the user.

The system uses six processing roles: 1) Urgent Warning; 2) Other Safety Sound; 3) Direct Speech; 4) User-Desired Content; 5) Environmental Context; 6) Unwanted Noise.

Such a design makes the model no longer limited to specific acoustic scenes and allows it to deal with new scenes that have not been encountered before.

3.4 Perception–Execution Interface

Since the descriptive language produced by the LALM cannot be directly used as a condition for audio processing, the system first uses a frozen CLAP Text Encoder to encode the description q_k of the k -th target sound into a 512-dimensional vector:

$$e_k = E_{\text{CLAP}}(q_k). \quad (3-2)$$

Different phrasing or vocabulary may express the same sound, but their text encodings in the embedding space are not completely identical. To reduce the inaccuracy of the following audio separation and processing caused by this shift, the system adds a Query-space Adapter between CLAP and the following execution layer, which separates and processes the audio. It is composed of a residual MLP and produces a correction to the original text vector:

$$e'_k = e_k + \alpha_k A_\theta(e_k), \quad (3-3)$$

where A_θ is the trainable residual mapping, α_k controls the correction strength, e'_k is the final embedding sent to the execution path.

During training, the last layer of the Adapter is initialized to zero. Therefore, at the beginning of training,

$$A_\theta(e_k) = 0, \quad e'_k = e_k. \quad (3-4)$$

In this way, the Adapter gradually learns a small correction.

3.5 Execution Path

The Execution Path performs the actual audio processing, including separating sounds with different roles from the audio, enhancing or suppressing them, and then mixing different sound components at certain proportions to produce the final output audio. It receives the current audio chunk from the ring buffer, as well as the text embeddings from the interface and the scene decisions from the LALM.

The Execution Path is divided into two steps:

1. Different sounds are first separated through the LASS model.
2. Audio processing then enhances, preserves, or attenuates the sounds according to the reasoning result of the large model.

3.5.1 Target Sound Extractor

The task of the Target Sound Extractor is to separate sound waveforms that can be independently adjusted from the current mixed audio according to the text embeddings provided by the interface. It uses a language-queried audio source separation model as its basic separator.

A single separation process of the Target Sound Extractor can be written as

$$\hat{s}_{k,t} = F_{\text{LASS}}(x_t, e'_{k,t}), \quad (3-5)$$

where x_t is the current mixed audio, $e'_{k,t}$ is the condition vector of the k -th target sound, $\hat{s}_{k,t}$ is the target sound estimated by LASS.

For a scene composed of multiple sound roles, the system executes LASS multiple times to obtain three independently controllable sound tracks, so that each sound can be controlled separately. Therefore, each scene containing multiple sounds has a sound-track set:

$$\hat{\mathcal{S}}_t = \{\hat{s}_{1,t}, \dots, \hat{s}_{K,t}\}. \quad (3-6)$$

Specifically, LASS first converts the one-dimensional mixed waveform into a complex time–frequency representation:

$$X_t = \text{STFT}(x_t). \quad (3-7)$$

This makes the periodic changes, harmonic structures, and energy variations of sounds more clearly represented.

The complex spectrogram then enters the ResUNet Encoder. The Encoder gradually compresses the time–frequency representation through multiple convolution and downsampling stages while increasing the number of feature channels. This structure can preserve the original acoustic information and reduce the difficulty of training a deep network.

The calibrated query embedding $e'_{k,t}$ is mapped into condition parameters for different network layers and is then injected into the intermediate features of ResUNet. In this way, the query is used to adjust the activation of different feature channels inside the network.

Since some sounds require a longer temporal range to be accurately identified, DPRNN is added to ResUNet. It first divides the longer intermediate feature sequence into multiple overlapping short chunks. It then separately learns the short-term frequency changes and transient structures inside each short chunk, as well as the long-term relationships between different chunks, such as whether a sound continues or repeats periodically.

The features then pass through the Decoder, which outputs three mask-related channels: 1) a magnitude mask, 2) a phase-direction real component, 3) a phase-direction imaginary component.

The magnitude mask passes through a sigmoid function and is used to estimate how much target sound should be preserved at each time–frequency position of the mixture. The other two outputs pass through a tanh function and are normalized into the phase-correction direction. The model therefore corrects both magnitude and phase to obtain the complex spectrum of the target sound. Finally, the target waveform is recovered through inverse STFT.

After amplitude calibration, the amplitude of each separated sound branch will not be very

different from its original contribution in the mixed audio.

3.5.2 Audio Processor

The task of the Audio Processor is to calculate enhancement or attenuation for different sound tracks according to the scene-reasoning result of the LALM, and then recombine multiple sound tracks with certain weights into a safe and audible output. It receives the sound tracks separated by LASS and finally outputs an integrated audible audio signal.

When processing audio, the Audio Processor considers the confidence of the LALM to perform either more active or more conservative processing:

$$r_t^{\text{eff}} = r_t (0.5 + 0.5c_t) . \quad (3-8)$$

When the confidence is high, the effective risk value is approximately equal to the risk value predicted by the model. When the confidence is low, the system adopts a more conservative processing strategy and does not immediately perform strong enhancement or suppression.

The Audio Processor adopts a gain strategy calibrated on the validation set. For each safety-related sound, the system first tests multiple candidate gain ceilings on the validation set and selects the minimum value that can sufficiently emphasize the target sound without excessively reducing the remaining background. During actual operation, the system scales the gain ceiling according to the current effective risk level. Therefore, low-risk sounds receive only limited enhancement, while the gains of high-risk sounds gradually approach the validated maximum value.

As the risk increases, safety-related sounds receive more noticeable enhancement. Speech directed toward the user is preserved or moderately enhanced according to its communication priority. Under ordinary conditions, user-desired content such as music and meaningful environmental sounds are preserved as much as possible. When a more urgent sound appears, these sounds can be moderately reduced, while irrelevant interfering sounds receive negative gains. Therefore, enhancement, preservation, and suppression are only different results of continuous gain control rather than three independent processing branches.

Finally, the Audio Processor applies the calculated gains to each extracted sound and the remaining scene audio, and then recombines them into the final output. This process adjusts the relative strength between different sounds. Therefore, the system can enhance safety sounds while preserving speech and user-desired content in the same scene:

$$y_t = g_{b,t}b_t + \sum_{k=1}^{K_t} g_{k,t}s_{k,t}, \quad (3-9)$$

where $s_{k,t}$ is the k -th extracted sound, $g_{k,t}$ is its corresponding gain, b_t is the remaining scene audio, $g_{b,t}$ is the gain of the remaining scene audio, y_t is the processed audio of the current chunk.

4 Experiments and Results

This section mainly evaluates the core designs of the system through two experiments. The first experiment studies whether the Query-space Adapter can correct language-query embeddings and improve the target-sound separation performance of LASS. This experiment compares Adapter V1 and Adapter V2 obtained through two stages of training, and reference methods including Wiener filtering and Base LASS without the Adapter. The second experiment studies the influence of audio-window length on dangerous-sound recognition and the following separation quality. The experiment is conducted with six types of dangerous sounds, ten types of interference, and ten chunk sizes, and selects a corresponding chunk size for each dangerous sound according to the recognition result, audio quality, and processing latency. In addition, this section further analyzes the effects and limitations of different system components through ablation experiments, false-alarm analysis, and a human questionnaire.

4.1 Experimental Setup

4.1.1 Adapter Training and Validation Data

The initial Adapter V1 mainly uses the FSD50K development data for training, in order to verify whether the Adapter can provide a basic improvement. However, the amount of available training audio for some sound categories is relatively small, and the distribution is not sufficiently balanced. Adapter V2 further combines audio data from FSD50K development, ESC-50, NIGENS, UrbanSound8K, and TUT Rare Sound Events.

After classifying the audio containing dangerous sounds, the dangerous-sound audio is mixed with interference audio under different signal-to-noise ratios to form the formal training and validation sets. This research constructs a training set containing 3,500 mixed-audio samples and a validation set containing 700 mixed-audio samples, covering seven common dangerous sounds in daily life, ten interference conditions, and five nominal signal-to-noise-ratio conditions in the manifests. During training, target-to-interference SNR is sampled continuously between -10 and 10 dB; validation uses the saved manifest values.

4.1.2 Danger-Sound Mixture Dataset

To analyze the influence of processing-window length on different dangerous sounds, this research constructs a danger-sound mixture dataset consisting of multiple data sources. The target sounds include Siren, Alarm, Crying and Sobbing, Screaming, Glass Breaking, and Bark. Each target-sound category is mixed with ten interference categories, including Crowd, Wind, Music, Engine, Speech, Rain, Fire, Footsteps, Knock, and Machinery. Five input signal-to-noise ratios of -10 , -5 , 0 , 5 , and 10 dB are used.

A total of 100 mixed-audio samples are generated for each dangerous sound, and the complete dataset therefore contains 600 mixtures. For each audio sample, the system separately tests ten chunk sizes of 2, 2.5, 3, 4, 5, 6, 7, 8, 9, and 10 seconds. Therefore, each danger-chunk condition contains the same 100 mixture-level units (6,000 repeated conditions in total); segmented processing can produce more than one record per mixture.

4.2 Baselines and Evaluation Metrics

The external comparison includes Unprocessed Mixture, the traditional Wiener Filter, Base LASS, and LASS with Adapter V2. Adapter V1 is evaluated in a separate historical ablation. GTCRN and DeepFilterNet3 are added in a separate environmental-target and speech-control comparison. Unprocessed Mixture represents the audio without any processing. The traditional Wiener Filter is a classical statistical signal-processing method established by Norbert Wiener during his work at MIT. Base LASS is the 32 kHz AudioSep-DP pretrained model released by Yin et al., whose authors are from Northwestern Polytechnical University, Alibaba Speech Lab, and Fortemedia Singapore. Adapter V1 and Adapter V2 are used to analyze the influence of different training data and calibration methods.

The objective evaluation mainly uses SNRi and SI-SDRi, with PESQ improvement and STOI improvement as supplementary indicators. SNRi and SI-SDRi measure the separation improvement of the processed sound relative to the input mixture. PESQ and STOI were originally designed for speech, and therefore they are only used as supplementary indicators for environmental sounds such as sirens, glass breaking, and dog barking. Systems are paired within each experiment; the external baseline, historical ablation, and later diagnostic runs must not be pooled. In the final evaluation questionnaire, 101 complete response records provide judgments of audio quality under different processing methods.

4.3 Main Experimental Results

4.3.1 Query-space Adapter Training and Evaluation

The Query-space Adapter experiment is used to study whether adding a text-vector correction module during audio-separation processing can improve language-queried audio source separation. Two natural-language descriptions may be semantically similar, but differences may still exist after they are encoded by the CLAP Text Encoder. Therefore, the Adapter is used here to correct this small difference and improve the audio-processing result.

The Adapter is a Multilayer Perceptron with a residual connection. For the original query embedding e_k , the Adapter outputs the corrected vector:

$$e'_k = e_k + \alpha_k A_\theta(e_k), \quad (4-1)$$

where A_θ is the trainable residual mapping and α_k controls the correction strength. The last layer of the Adapter is initialized to zero, and therefore $e'_k = e_k$ at the beginning of training.

The training loss is

$$\mathcal{L} = \mathcal{L}_{\text{SI-SDR}} + 0.5\mathcal{L}_{\text{L1}} + 0.1\mathcal{L}_{\text{MRSTFT}} + 0.01\|e'_k - e_k\|_2. \quad (4-2)$$

Here, $\mathcal{L}_{\text{SI-SDR}}$ represents the negative SI-SDR loss for improving the overall separation quality, $\mathcal{L}_{\text{L1}}$ represents the waveform difference between the separated audio and the clean target audio, and $\mathcal{L}_{\text{MRSTFT}}$ represents their spectral difference under multiple time–frequency resolutions. The final term limits the distance between the adapted and original query embeddings to prevent excessive correction.

During training, the learning rate is 10^{-4} , and the weight decay is 10^{-4} . After each epoch, SNRi and SI-SDRi are calculated on the validation set, and the best weights are finally saved according to the validation SI-SDRi.

Adapter V1 and Adapter V2 use the same MLP structure, and their main differences are the training data and validation method. V1 contains approximately 1,000 training samples and 200 validation samples, while V2 contains 3,500 training samples and 700 validation samples, with a more balanced sampling distribution. Both versions use an Adapter strength $\alpha_k \in [0, 1]$, which controls how much of the learned residual correction is applied; the historical versions use different strength recommendations.

On the 600 paired samples in the external baseline run, Base LASS has source-weighted mean SNRi and SI-SDRi of 12.77 and 11.68 dB, while Adapter V2 reaches 13.73 and 12.69 dB. The paired gains are +0.957 dB and +1.015 dB, with 95% confidence intervals of [0.557, 1.369]

and [0.543, 1.511], respectively. PESQ improves by 0.085 over 547 valid pairs, with a 95% interval of [0.048, 0.121]; the STOI difference is not significant. Table 4.1 and these differences are calculated from the same per-sample file and the same source weighting. Values rounded for display need not subtract exactly.

Table 4.1: External comparison reconstructed from one version of the 600-mixture results. Entries are target-source means with source-bootstrap 95% intervals. SNRi, SI-SDRi and STOI have 600 valid observations; PESQ has 547. Adapter V1 is not included because it was evaluated in a different run.

System	SNRi (dB)	SI-SDRi (dB)	Δ PESQ	Δ STOI
Unprocessed Mixture	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.000 [0.000, 0.000]
Wiener Filter	-0.95 [-1.96, -0.12]	-1.06 [-2.07, -0.19]	-0.01 [-0.03, -0.00]	-0.002 [-0.003, -0.002]
Base LASS	12.77 [10.95, 14.49]	11.68 [9.75, 13.45]	0.89 [0.81, 0.98]	0.034 [0.022, 0.045]
LASS + Adapter V2	13.73 [11.95, 15.41]	12.69 [10.81, 14.45]	0.98 [0.90, 1.07]	0.037 [0.024, 0.049]

Different sound categories do not obtain the same improvement from the Adapter. Glass Breaking and Screaming show the most obvious SI-SDRi gains, which are approximately 2.00 dB and 2.43 dB, respectively, while the improvements for Siren, Bark, and Crying and Sobbing are relatively smaller. This indicates that the effect of query correction is related to the sound category itself, and the observed gains vary by category; this comparison does not establish the cause of those differences.

4.3.2 Chunk-size Evaluation and Adaptive Selection

The chunk-size experiment is used to study how the length of the LALM audio-analysis window influences dangerous-sound recognition and the final audio-processing result. When the LALM understands a sound event, a shorter chunk size may provide a faster response, but it may contain only part of the sound event. Therefore, the chunk-size experiment aims to find an optimal scheme that jointly considers audio quality, sound recognition, and response latency.

SNRi, SI-SDRi, PESQ, and STOI have different numerical ranges and therefore cannot be directly added together. The experiment first normalizes the four indicators under the ten chunk sizes within each dangerous-sound category.

To reduce the influence of random fluctuation on the average value, the 95% lower confidence bound is used for each indicator:

$$m_{\text{LCB}} = \bar{m} - 1.96 \frac{s_m}{\sqrt{n_m}}, \quad (4-3)$$

where $\bar{m}$ represents the mean value of the indicator, s_m represents the sample standard deviation, and n_m represents the number of valid samples for the indicator. This method considers not only the average performance but also whether the result is stable. When the mean values of two chunk sizes are close, the setting with a smaller standard deviation obtains a higher lower confidence bound.

The four indicators are then combined into an audio-quality score:

$$Q_y(c) = 0.25N_y(\text{SNRi}_{\text{LCB}}) + 0.35N_y(\text{SI-SDRi}_{\text{LCB}}) + 0.20N_y(\text{PESQ}_{\text{LCB}}) + 0.20N_y(\text{STOI}_{\text{LCB}}), \quad (4-4)$$

where y represents the dangerous-sound category, c represents the chunk size, and $N_y(\cdot)$ represents min-max normalization among the ten chunk sizes of the same dangerous sound.

SNRi represents the improvement in general signal quality, SI-SDRi represents the separation quality without being affected by overall amplitude scaling, PESQ represents perceptual quality, and STOI represents sound intelligibility. SI-SDRi is assigned the highest weight because target-sound separation quality is the main audio indicator in this experiment. SNRi and SI-SDRi are improvement values relative to the input mixture, while PESQ and STOI are direct reference-based scores. After min-max normalization, a higher value represents better relative performance among the ten chunk sizes of the same dangerous sound. The system selects the largest Combined Performance Score, calculated as $0.55Q_y(c) + 0.35R_y(c) + 0.10L_y(c)$, where R_y and L_y are the saved normalized detection and latency scores. This is an exploratory selection on the same evaluated data.

If only the audio-quality score composed of SNRi, SI-SDRi, PESQ, and STOI is considered, all six dangerous-sound categories select 10 seconds. This indicates that a longer window usually provides more complete acoustic information and produces higher offline separation quality. However, this result does not consider how long the system needs to wait before obtaining the recognition result.

When detection rate and decision latency are added to the Combined Performance Score, the selected results for different dangerous sounds become different. The Combined Performance Scores of the different dangerous signals under their corresponding chunk sizes are shown in Figure 4.1. The chunk sizes that has the greatest score for each danger type are framed by red frames.

The figure compares relative window scores within the evaluated system, rather than di-

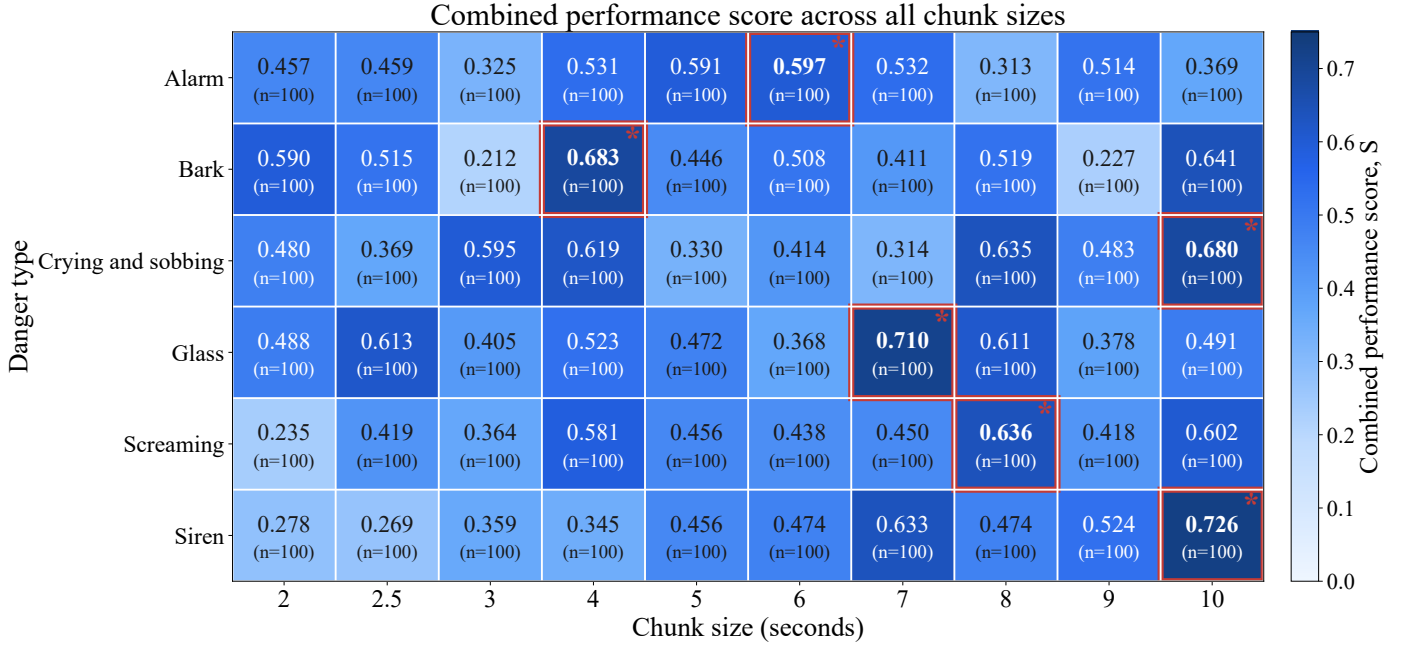


Figure 4.1: Combined Performance Scores of different dangerous-sound categories under different chunk sizes. Each danger–chunk condition contains at least 100 experimental samples.

rectly establishing superiority over Base LASS. Specifically, Bark obtains the highest score at 4 seconds because dog barking usually has relatively clear short-term characteristics. Alarm, Glass Breaking, and Screaming select 6, 7, and 8 seconds, respectively, while Crying and Sobbing and Siren still select 10 seconds because crying and sirens may be continuous and repetitive, and a longer window can present more complete sound characteristics. This indicates that different sounds do not have the same requirement for temporal context, but the selected policy has not yet been tested against fixed windows on an independent held-out set.

After obtaining the better chunk size corresponding to each danger type, this research designs an adaptive chunk-size module. The system first uses a default window of 10 seconds. When the LALM analyzes the current audio and identifies one of the dangerous sounds that has been studied, the corresponding recommended chunk size is applied to the next processing window, as a proposed adaptation rule. Its benefit on new mixtures and scene transitions remains to be evaluated.

4.3.3 Deep Denoising Baselines and Speech Control

To compare language-queried separation with deep speech denoising, this research additionally evaluates GTCRN [16] and DeepFilterNet3 [18]. AudioSep-DP is the frozen separator used in SmartHearing; this comparison uses the standard target query without the Adapter or subsequent risk-based mixing. It is a separator comparison, not a comparison of the complete SmartHearing system.

Figure 4.2 reports 550 input-valid environmental-target mixtures and a separate control of 100 speech-target mixtures from 20 target recordings. On environmental targets, AudioSep-DP obtains SNRi/SI-SDRi of 12.35/11.16 dB, compared with $-2.25/-18.11$ dB for GTCRN and $-3.39/-25.46$ dB for DeepFilterNet3. Speech denoisers can suppress non-speech warning sounds because those sounds are not their intended target; these negative gains do not establish that the models are ineffective at speech enhancement.

In the speech control, SNRi/SI-SDRi are 16.86/16.53 dB for AudioSep-DP, 11.30/10.79 dB for GTCRN, and 12.37/12.24 dB for DeepFilterNet3. Both denoisers obtain positive gains, supporting this distinction between speech enhancement and environmental-target extraction. The small speech control does not replace an independent standard speech benchmark. These later 550-pair results are separate from the 600-mixture external comparison above.

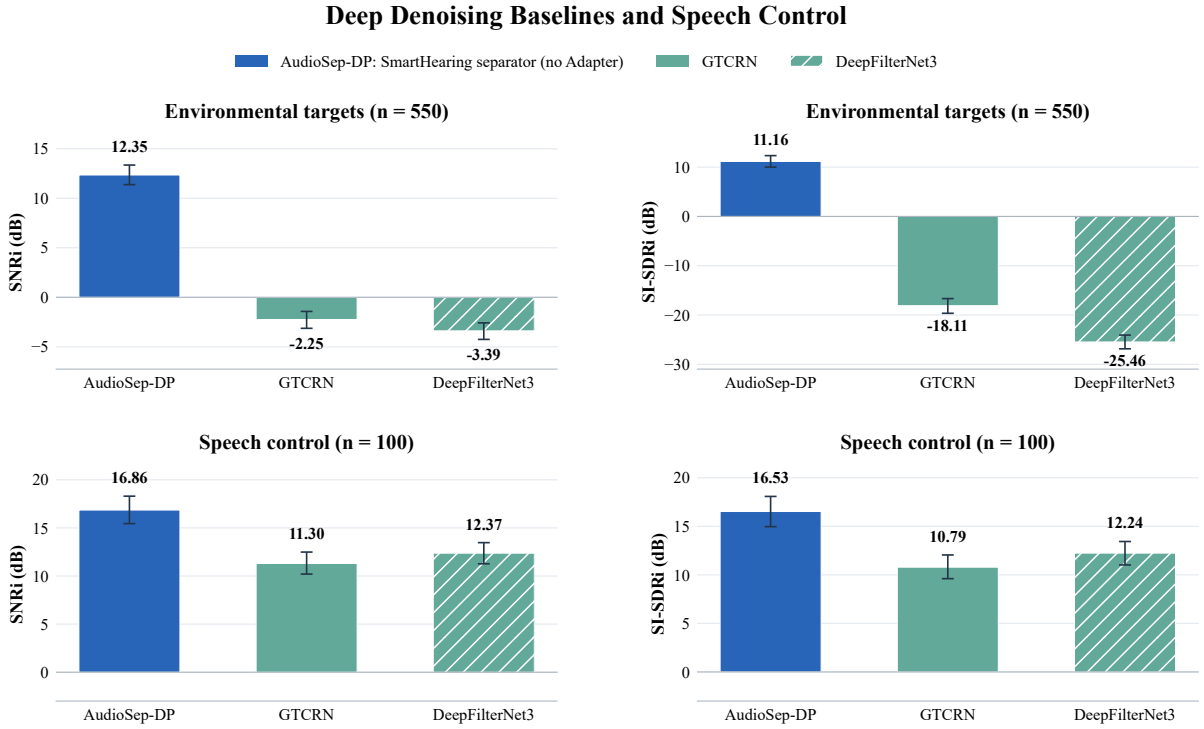


Figure 4.2: Deep denoising baselines and speech control. AudioSep-DP is the SmartHearing separator without the Adapter. Bars show mixture-level means; error bars are 95% target-source cluster-bootstrap intervals (5,000 resamples). Environmental and speech tasks use 550 and 100 mixtures, respectively.

4.4 Ablation and Further Analyses

In this section, this study further separates the key components of the system, analyzes which parts contribute to the performance improvement, and discusses how different design choices should be balanced. Specifically, this study compares two versions of the Query-space Adapter, analyzes where and why the LALM produces errors, and tests different sound-

processing settings. The cross-dataset performance of the system is then evaluated through experiments. Finally, a listening questionnaire is used to study people’s subjective evaluations of the processed audio quality.

4.4.1 Adapter V1 and V2 Ablation

This study designed one control group and two experimental groups. The control group was Base LASS, while the two experimental groups were Base LASS with Adapter V1 and Base LASS with Adapter V2. These three groups were compared on 600 identifiers within the historical ablation run. This run is distinct from Table 4.1: 467 recorded input SNRs differ between the two runs, and the correction-strength settings also differ. Its gains are therefore retained separately rather than substituted into the external comparison.

Table 4.2: Historical within-run paired differences among Base LASS, Adapter V1, and Adapter V2. Values are source-weighted mean differences with 95% intervals. SNRi, SI-SDRi and STOI use 600 pairs; PESQ uses 547. Inputs and strength settings differ from Table 4.1.

Comparison	SNRi (dB)	SI-SDRi (dB)	Δ PESQ	Δ STOI
Adapter V1 – Base	+0.362 [0.155, 0.568]	+0.393 [0.025, 0.761]	+0.033 [0.013, 0.052]	–0.000 [–0.003, 0.003]
Adapter V2 – Base	+0.970 [0.598, 1.341]	+1.138 [0.637, 1.639]	+0.073 [0.036, 0.110]	+0.004 [–0.002, 0.010]
Adapter V2 – Adapter V1	+0.608 [0.254, 0.962]	+0.745 [0.274, 1.216]	+0.040 [0.004, 0.076]	+0.004 [–0.002, 0.010]

The results show that Adapter V1 already produces a small improvement over the strong Base LASS baseline, proving that the adapter can improve the model performance. After the dataset was further expanded and balanced, Adapter V2 significantly outperformed Adapter V1 in SNRi, SI-SDRi, and PESQ. This shows that V2 effectively improves the overall audio-processing quality of the model. However, the confidence interval of STOI still crosses zero. Therefore, the adapter does not stably improve speech intelligibility.

4.4.2 Comparison of LALMs

Comparison Results This research also conducted an LALM comparison experiment between Qwen3.5-Omni and Gemini 3.5 Flash. The two models separately generated sound-processing instructions from the input audio, which was then processed by the same LASS separator. Therefore, this experiment allows the analysis to focus on how the choice of LALM affects subsequent processing. The same frozen Base LASS was used on both sides, without the Adapter or subsequent risk-based gain adjustment and mixing.

This research used a total of 700 paired audio samples to compare the recognition accuracy and false-alarm rates of the two LALMs, and used quantitative metrics to compare their effects on audio-processing quality. The comparison included 600 target-present samples with five-second inputs and 100 target-absent samples with three-second inputs. The target-present samples were also used for the downstream audio-quality comparison.

The target-sound recognition rate examines whether the model recognizes the target sound in its reasoning, while the correct extraction instruction rate examines whether the model finally instructs LASS to separate that sound. As shown in Table 4.3, Qwen’s recognition rate is 3.83 percentage points higher than Gemini’s, but its correct extraction instruction rate is only 0.17 percentage points higher. This shows that recognizing the target sound sometimes does not mean that the LALM can correctly convert this recognition into an execution instruction for LASS.

Table 4.3: Recognition and false-alarm results of the two LALMs. Holm correction is applied across the five reported outcomes.

Metric	n	Qwen	Gemini	p_{Holm}
Target-Sound Recognition Rate	600	76.67%	72.83%	0.0823
Correct Extraction Instruction Rate	600	68.17%	68.00%	1.0000
Extraction Instruction False-Alarm Rate	100	15.00%	18.00%	1.0000
High-Risk False-Alarm Rate	100	22.00%	7.00%	0.0036
Combined False-Alarm Rate	100	22.00%	18.00%	1.0000

The Extraction Instruction False-Alarm Rate measures how often the model outputs a target query and unnecessarily activates the separator on a target-absent sample. The High-Risk False-Alarm Rate measures how often the model assigns such a sample a global danger score of at least 0.8. The Combined False-Alarm Rate counts a sample if it produces an extraction instruction false alarm, a high-risk false alarm, or a primary-safety sound role. These outcomes can overlap.

In the false-alarm analysis, Qwen has a higher high-risk false-alarm rate than Gemini, but a lower extraction instruction false-alarm rate. The difference in high-risk false alarms remains statistically significant after Holm correction ($p_{\text{Holm}} = 0.0036$), while the differences in extraction instruction false alarms and combined false alarms are not significant. Therefore, Gemini produces fewer high-risk false alarms under the current threshold, but this does not mean that it performs better on every false-alarm measure. Differences in the risk-score distributions of the two models may also affect the frequency of false alarms.

Figure 4.3 shows the four quantitative metrics for audio processed using the two LALMs, together with their recorded API response times.

Among the 600 target-present evaluation windows, 50 contained a mixture identical to the target reference, making the input SNR and SI-SDR baselines degenerate. All 600 results were

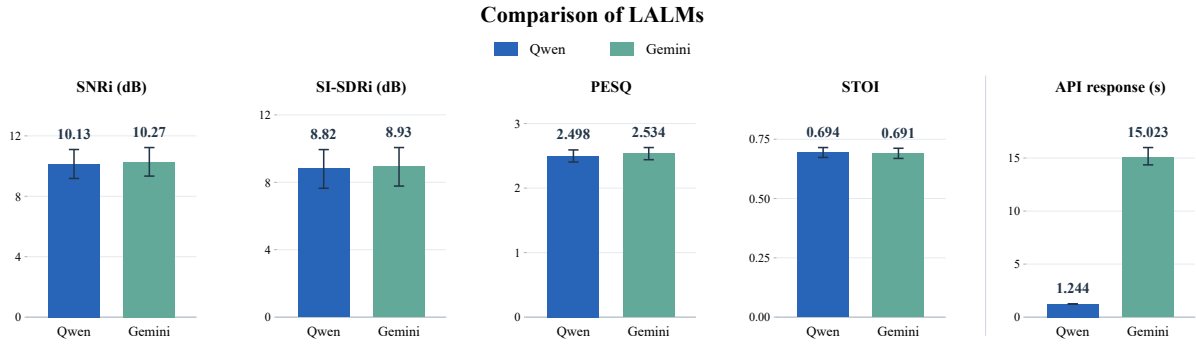


Figure 4.3: Audio-quality and API response-time comparison. The four audio-quality panels compare Qwen + LASS with Gemini + LASS using the same separator. Error bars indicate 95% bootstrap confidence intervals: means for audio quality (550 valid SNRi/SI-SDRi pairs and 527 PESQ/STOI pairs) and medians for response time (545 matched, explicitly serial target-present requests).

retained, while the figure shows a post-hoc diagnostic analysis of the remaining 550 input-valid pairs. This selection was based on the input audio rather than the relative performance of the two models.

As shown in the figure, Gemini has slightly higher SNRi, SI-SDRi, and PESQ values, while Qwen has a slightly higher STOI value. However, the confidence intervals for the overall differences in all four metrics cross zero, and none of the four differences is significant after Holm correction for multiple comparisons. Therefore, these slight differences cannot prove that either Qwen or Gemini has better overall audio quality. No statistically significant overall audio-quality difference was detected, but this does not establish equivalence between the two models.

However, it is worth noting that Qwen’s recorded response time is lower than Gemini’s, with a median of 1.244 seconds compared with Gemini’s 15.023 seconds on 545 matched target-present requests explicitly recorded as serial. This timing subset is not the 550-pair audio-quality subset. Therefore, Qwen had a shorter observed API response time under the current deployment conditions. The timing boundaries are not identical: Qwen timing begins after the WebSocket connection is established, while Gemini timing begins before the HTTP request. Network and deployment conditions also differ. These measurements should therefore not be interpreted as a controlled comparison of pure model inference speed or as the end-to-end latency of the hearing assistance system.

Further Analysis of False Alarms The comparison above shows that false alarms remain an issue for both LALMs. Following this comparison, the previous Qwen-only analysis is retained to examine whether increasing the danger-level threshold can reduce false system activation.

This earlier analysis uses a fixed three-second chunk and compares three decision settings:

no danger-level threshold, a validation-selected threshold of 0.60, and a stricter threshold of 0.90. It is a separate post-hoc exploratory analysis, rather than a further comparison between Qwen and Gemini. The calibration subset contains 301 target-present and 48 target-absent samples, while the test subset contains 299 target-present and 52 target-absent samples. The threshold of 0.60 was selected on the calibration subset, and the results in Table 4.4 are reported on the test subset.

Here, the threshold determines whether an extraction instruction is allowed to activate the system. Correct activation requires both the correct target query and a danger score meeting the threshold. False activation occurs when a target-absent sample produces a non-no_target query and meets the threshold. This differs from the high-risk false-alarm measure above, which uses a fixed score threshold of 0.8.

Table 4.4: Influence of the danger-level threshold on correct and false system activation in the previous Qwen-only analysis. Results use 299 target-present and 52 target-absent test samples.

Decision setting	Correct Danger-Response Rate	False Danger-Response Rate	Balanced Accuracy
No Danger-Level Threshold	0.679	0.135	0.772
Validation-Selected Threshold (0.60)	0.679	0.135	0.772
Strict Threshold (0.90)	0.264	0.019	0.622

The results show that the threshold of 0.60 produces exactly the same result as using no danger-level threshold, and therefore does not provide an additional improvement. When the threshold is increased to 0.90, the false danger-response rate decreases from 0.135 to 0.019, but the correct danger-response rate also decreases from 0.679 to 0.264. Therefore, in this experiment, increasing a unified danger threshold does not resolve both missed detections and false alarms. A possible next step is to further improve the accuracy and consistency of the LALM through task-specific fine-tuning.

4.4.3 Cross-Dataset Generalization Analysis

To examine the generalization performance of the adapter, this study further uses ESC-50 as a dataset-held-out evaluation source. A leave-one-dataset-out Adapter was trained and validated without using ESC-50-labelled target or interference samples, and was then tested on 145 ESC-50-labelled test mixtures. The differences between this adapter and Base LASS are compared on these samples.

The confidence intervals of SNR_i and SI-SDR_i are both entirely above zero. This supports improved separation under dataset-label exclusion. It does not establish recording-level independence across catalogs or absence of pretraining exposure. However, the confidence in-

Table 4.5: Cross-dataset performance of the leave-one-dataset-out Adapter on ESC-50-labelled test mixtures; recording-level cross-catalog independence is not established. Values are Adapter minus Base LASS.

Metric	Valid n	Mean difference	95% CI
SNRi (dB)	145	+1.323	[0.499, 2.172]
SI-SDRi (dB)	145	+1.192	[0.248, 2.146]
Δ PESQ	134	+0.064	[−0.031, 0.148]
Δ STOI	145	+0.002	[−0.012, 0.013]

tervals of PESQ and STOI both cross zero. Therefore, not all evaluation metrics achieve stable cross-dataset improvement, and the perceptual quality and speech intelligibility of the processed audio still have room for improvement.

4.4.4 Runtime Analysis

The runtime experiment was conducted on 600 five-second mixtures. The local processing times of the Wiener Filter, Base LASS, and LASS with Adapter V2 were separately recorded.

Table 4.6: Local runtime recomputed from the same 600-sample external per-case result file used for Table 4.1; medians and median runtime/5 are reported.

System	Number of samples	Median runtime (s)	Median RTF
Wiener Filter	600	0.0131	0.0026
Base LASS	600	0.0664	0.0133
LASS + Adapter V2	600	0.0671	0.0134

The real-time factor, or RTF, is defined as the processing time divided by the duration of the input audio. An RTF below 1 means that the model processes the audio faster than the audio is played. The median RTF of Adapter V2 is 0.0134, while that of Base LASS is 0.0133, with a difference of only approximately 0.0001. This shows that the adapter introduces almost no additional local computational cost. At the same time, the experiment supports the feasibility of low-latency local audio processing. This excludes window acquisition, queueing, cloud communication and inference, and playback buffering. Complete response must be measured from event onset to the first played output reflecting the new decision; separate medians cannot simply be added because stages may overlap. Such end-to-end measurements have not been completed.

4.5 Case Study

4.5.1 Spectrogram Comparison

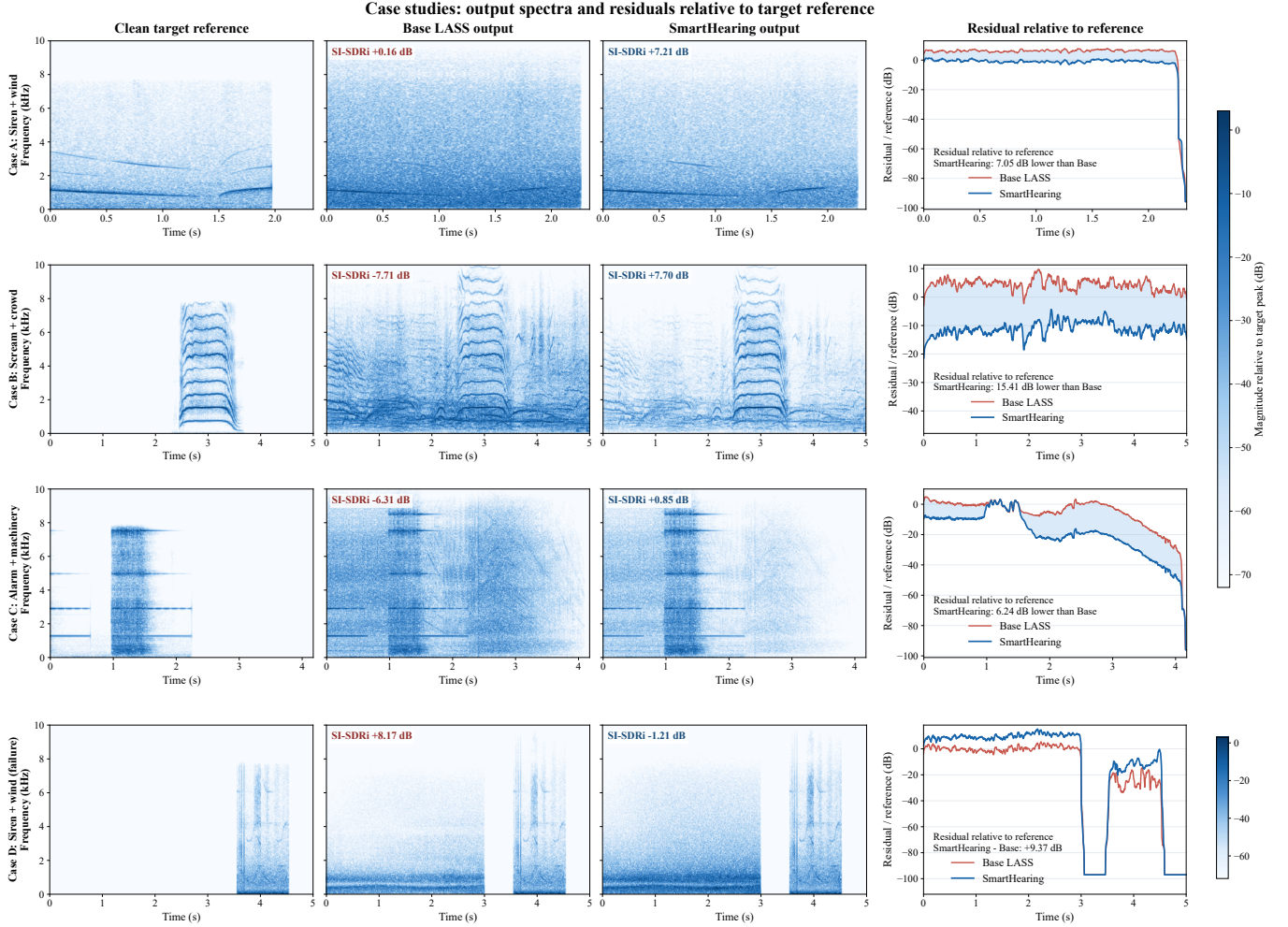


Figure 4.4: Target reference spectrograms, scale-aligned output spectrograms, and residual-energy curves: (A) siren with wind; (B) screaming with crowd; (C) alarm with machinery; (D) a siren-wind failure. The cases use the same historical separation-and-rendering protocol and are selected illustrations, not a random sample. Colours within each case use a shared reference peak.

Case A shows the processing result of a siren under wind-noise interference. The siren spectrum has continuously changing frequency stripes, with the main component at around 1,000 Hz. Its frequency first decreases and then increases over time, as shown in the leftmost image of the first group in Fig.4.4. Base LASS keeps part of the frequency change of the siren, but there are still many dense noise points in the output. SmartHearing also keeps the main spectrum of the siren and further strengthens it, while reducing part of the background noise. As a result, the intensity difference between the target and non-target components becomes larger, making the target sound easier to hear. The residual plot further shows that the residual curve of SmartHearing stays below that of Base LASS, and its residual energy is about 7.05 dB lower than that of

Base LASS. This means that the target sound is closer to the reference sound. In the quantitative evaluation, SI-SDRi increases from 0.160 dB for Base LASS to 7.208 dB for SmartHearing, an increase of 7.048 dB, while SNRi increases by 2.058 dB. These quantitative results are consistent with what can be seen in the spectrogram, where the background components are reduced and the siren stripes are strengthened. However, SmartHearing does not remove all of the background sounds. The spectrogram alone cannot distinguish intended background preservation from separation error, or establish that conversations have been preserved.

Case B shows the processing result of screaming under crowd-noise interference. The output of Base LASS can show the main harmonics of the screaming, but there is still clear background noise and some extra frequency stripes, especially in the low-frequency region and in the periods before and after the screaming appears. SmartHearing successfully reproduces the main frequency positions and harmonic changes of the screaming while reducing the background noise, making the target more obvious. The residual curve shows that the residual of SmartHearing is lower than that of Base LASS throughout the whole time period, with the residual energy reduced by about 15.41 dB. For the whole segment, SI-SDRi increases by 15.409 dB and SNRi increases by 3.185 dB. However, some extra stripes can still be seen in the output, which means that interference from the crowd still remains. This case is important because both the target screaming and the crowd interference contain human-voice-related components. In this sample, SmartHearing can selectively process the specified screaming sound, showing its ability to process a specific sound target.

Case C shows the processing result of an alarm under mechanical-noise interference. In the target reference spectrum, horizontal frequency stripes can be seen at around 1,300 Hz and 2,900 Hz, together with a wider-band component appearing at around 1 second. From the result, Base LASS keeps the main horizontal stripes of the alarm, but there are also extra components covering a relatively large range of time and frequency. SmartHearing also keeps the main frequency positions of the alarm while reducing part of these extra components. Lighter regions can be seen in both the spectrogram and the residual plot of SmartHearing. The residual plot also shows that, for most of the time, the residual of SmartHearing is lower than that of Base LASS, although the two residual curves become close between 0.9 and 1.5 seconds. This shows that SmartHearing still has some difference from the reference sound in more complex local regions. Quantitatively, the overall residual energy of SmartHearing is reduced by about 6.24 dB. SI-SDRi increases by 7.162 dB and SNRi increases by 5.901 dB, showing that SmartHearing improves the overall relationship between the output audio and the target reference, although its processing is still not sufficient in some more complex local regions.

Failure analysis Case D is a siren–wind mixture (danger_00012, segment 1, corresponding to 5–10 s of the original recording). Base LASS achieves an SI-SDRi of 8.165 dB, while SmartHearing achieves –1.209 dB, a decrease of 9.374 dB. SNRi decreases from 6.189 to –1.103 dB. The target reference is nearly silent during the first approximately three seconds of this segment, but both outputs retain low-frequency and broadband energy; this residual is stronger for SmartHearing. Its scale-aligned residual energy is 9.37 dB higher than Base. Both runs use the same primary query, siren, the same secondary query, car horn, and identical nominal rendering gains. The loss therefore cannot be attributed to a difference in primary target selection. It indicates a downstream extraction-and-rendering failure under the adapted configuration, but does not isolate the contributions of individual stems and their mixing. Component-level ablations would be needed to determine the mechanism. Case D is the largest SI-SDRi loss among 343 eligible, correctly routed, target-active five-second pairs and is shown for diagnosis, not as an estimate of the overall failure rate.

4.5.2 Human Listening Evaluation

Table 4.7: Subjective Evaluation Results of SmartHearing and Baseline Conditions

Metric	Original Mixture	Base LASS	SmartHearing
Target-sound detectability (%)	21.5	28.1	50.5
Final preference (%)	23.4	27.1	49.5
Mean overall subjective score	5.409	5.337	5.198

This study also released a questionnaire to evaluate people’s subjective perception of the audio processed by SmartHearing and the baseline model. The current questionnaire export contains 101 responses. Each participant listened to three scenes, producing 303 detectability and final-preference choices in total.

From the collected data listed in Table.4.7. SmartHearing shows an obvious advantage in the target-sound detection task. Among the 303 choices from the three scenes, SmartHearing was selected as the condition in which the target sound was easiest to notice in 50.5% of the cases. This proportion was higher than 28.1% for Base LASS and 21.5% for the original unprocessed mixture. These response records favour the processed condition for target salience, but do not isolate the Adapter from the listening-specific gain policy. Recruitment and listening compliance have not been independently verified, so these are preliminary questionnaire-record results rather than validated clinical evidence.

SmartHearing was also selected as the final preferred condition in 49.5% of the choices, compared with 27.1% for Base LASS and 23.4% for the original mixture. This shows that the

choice-based results favor the audio processed by SmartHearing.

However, the mean overall subjective scores of the original mixture, Base LASS, and SmartHearing were 5.409, 5.337, and 5.198, respectively. The difference among the three scores did not reach statistical significance in the Friedman test ($p = 0.0738$). This shows that although SmartHearing increases the prominence of the target sound in the choice-based task, it does not yet produce a significant improvement in the overall subjective score. The processing may change the naturalness of the background sound and the listening comfort, and therefore the overall listening quality still needs to be further improved.

5 Conclusion

This research addresses the problem that conventional hearing aid systems mainly focus on speech enhancement or overall noise reduction and therefore have difficulty distinguishing sounds with different levels of importance in complex acoustic scenes, and proposes the SmartHearing personalized intelligent hearing assistance system. The system is divided into a perception path and an execution path. In the perception path, the LALM first analyzes the sound objects, roles, danger levels, and confidence levels in the scene. The target-sound descriptions are then converted into text embeddings, which are corrected by the Adapter. In the execution path, the LASS model separates the audio according to the text embeddings, while the Audio Processor finally enhances, preserves, or attenuates different audio tracks according to their sound roles and danger levels. The system also adaptively adjusts the chunk size for the next processing cycle according to the current scene, thereby achieving a balance between sound quality and response speed.

The experimental results show that adding an Adapter between the perception and execution paths can improve the separation quality of target sounds. In the external baseline experiment, compared with the Base LASS model proposed by researchers from Northwestern Polytechnical University, Alibaba Speech Lab, and Fortemedia Singapore, Adapter V2 improved SNRi and SI-SDRi by 0.957 dB and 1.015 dB, respectively, while PESQ was improved by 0.085. The separate historical V1/V2 run also favoured V2, but used different input and correction-strength settings from the external comparison. In the Cross-Dataset Generalization Analysis, the Adapter still improved SNRi and SI-SDRi by 1.323 dB and 1.192 dB, respectively, on the ESC-50 test data, a dataset label excluded from the separate Adapter training and validation manifests; original-recording independence across catalogs is not established. This result indicates that the Adapter has a certain degree of generalizability and can extend its performance to an unseen dataset source.

The danger–chunk experiment further shows that there is no single optimal chunk size suitable for all sounds. When sound recognition quality, danger level, and processing time were considered together, the recommended windows for different dangerous sounds were mainly concentrated between 4 and 8 seconds, while a 10-second window was also selected in some cases. The selection of different chunk sizes is related to the duration and periodicity of the sounds. These scores describe exploratory window selection on the same mixtures and do not establish held-out superiority over fixed windows.

Meanwhile, in the latency test, the model showed a relatively low local processing latency. The asynchronous processing of the perception and execution paths further provides a reasonable basis for real-time audio processing.

In the final human listening questionnaire, the audio processed by Adapter V2 received a user selection rate of 50.5%, for the target-easiest-to-notice question, rather than overall satisfaction, compared with the initial unprocessed mixture, which received 21.5%, and the audio processed by the Base LASS model, which received 28.1%.

However, one limitation of the current system is that the Correct Extraction Query Rate of the LALM has not yet reached a fully reliable level, resulting in missed detections or false alarms. Therefore, future work will first improve the reliability of the connection between the perception and execution paths, including introducing cross-checking of the LALM outputs to improve decision confidence. Meanwhile, the system still needs to be extended to a more open set of sounds. When conditions permit, real hearing-impaired users should also be invited to participate in the evaluation.

6 Acknowledgements

The idea for this project originated from my interest in listening to classical music. My father has also collected antiques, so we have an old gramophone at home. When I tried using this gramophone, I noticed that the audio it played had a noticeable sense of muddiness. Therefore, I worked with another student on a signal-processing-based project investigating audio quality restoration for gramophones. In this project, we compared the performance of firwin2 with a dynamic spectral restoration algorithm that we developed ourselves. We ultimately found that our algorithm could improve the audio quality by approximately 37

At school, I participated in an artificial intelligence elective course. In the course, I systematically studied and practiced classic models including MLP, CNN, RNN, and Transformer, which also provided a foundation for me to develop this project. After passing the final interview for the school elective course, I was given the opportunity to carry out an individual artificial-intelligence-related project for more than half a year. Through this opportunity, I met Mr. Kong and invited him to serve as my supervisor. After analysing the foundational knowledge I had already acquired and the direction in which I wanted to develop my project, I decided to further explore the field of audio. Because my grandfather had experienced hearing impairment, I then thought of starting with hearing aids and researching a new type of hearing-aid algorithm, with the aim of improving the quality of life of people with hearing impairment and elderly people.

In this project, I completed all of the work involving the literature review, methodology, model design, programming, dataset construction, experiments, data processing, analysis, figure production, and paper writing. Mr. Kong is a teacher from the Institute of Artificial Intelligence at Peking University who participates in the follow-up project programme of our school's artificial intelligence elective course, and his supervision was provided without charge. The way Mr. Kong and I worked together was that, during the early stage of the project, we held a biweekly update meeting every two weeks. During each meeting, I reported to him on the work I had completed during the previous two weeks. Based on my work, he would give me suggestions for improvement and advise me on which key area I should focus on next. At the same time, when we encountered certain difficulties, we would also think together about possible ways to solve them, which will be discussed in detail later.

Mr. Kong's main role in the project was to evaluate the work I had completed with me during each meeting, identify its shortcomings, and help me address them. In terms of paper writing, after I independently completed the entire first draft, Mr. Kong gave me suggestions for revision, and I made a second and third round of revisions based on these suggestions. Specific suggestions he made included increasing the amount of data, adding baseline comparisons with

other large models, experimenting with modifications to the prompt to reduce the large model's bias toward human speech, and regularly backing up my work logs. In terms of paper writing, he also suggested that I revise some aspects of the text formatting and remove the small notes underneath figures. The supervisor did not modify my research design and did not directly participate in revising the paper.

The technical support included an RTX 5060 GPU that I purchased myself.

One research difficulty was that, when searching for datasets, there was no readily available dataset that both simulated realistic urban life and contained clear labels. After downloading multiple datasets, I decided to mix selected audio samples from the FSD50K development data to create mixture audio containing both target and interfering sounds, which was used as the final dataset for training the model. There were also many other technical problems during the research process, including issues involving model interfaces and interrupted network downloads, but I ultimately overcame all of these technical problems.

I have used AI several times throughout this paper because English is not my first language. The use was mainly in cases when there was uncertainty of several phrases and grammars when writing the essay. Also, AI generative model (GPT 5.6) is used to revise the code fully based on my instruction, as their sometimes are errors during code running. They were not used to determine the research question, design the methodology, conduct experiments, generate or alter experimental data, perform the core data analysis, or draw scientific conclusions.

Research question, study design, logic of the code, datasets, experiments, data analysis are all performed by me. The language assistance AI response was not directly copied into the paper. I reviewed and revised all AI-assisted language suggestions before incorporating them into the final manuscript. For transparency, links to the relevant AI chat records are provided below.

<https://chatgpt.com/share/6a5b134f-e860-83ee-945e-2bc6ac683560>

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Appendix

This appendix gives further details about the datasets, model training, experimental settings, listening questionnaire, and code used in this research. These details are placed here so that the main experiments can be understood and reproduced without adding too many implementation details to the main paper. The raw audio files are not redistributed because they are still covered by the licences of the original datasets. Private API keys and file paths from the local computer are not included.

Experimental Materials

Three types of materials were prepared for the main experiments: the data used to train the Adapter, the data used in the chunk-size experiment, and the audio used in the listening questionnaire. The Adapter experiment tests whether correcting the text-query embedding can improve language-queried sound separation. The chunk-size experiment examines whether different safety-related sounds require different lengths of audio context. The listening questionnaire asks whether the processed target sounds are easier for listeners to notice.

Table 6.1: Summary of the main experimental materials.

Material	Scale	Use in the experiment
Adapter V1 data	1,000 training and 200 validation mixtures	Initial small-scale training using the clean target sounds available in the FSD50K development data.
Adapter V2 data	3,500 training and 700 validation mixtures	Adapter training using seven target categories, audio from several datasets, and more balanced interference conditions.
Chunk-size experiment	600 mixtures and 6,000 test cases	Comparison of ten chunk sizes for six representative safety-related sounds.
LALM test samples	600 target-present and 100 target-absent samples	Paired LALM comparison; the downstream quality diagnostic retains 550 input-valid target-present pairs.
ESC-50 generalization test	145 mixtures	Evaluation on ESC-50 using an Adapter trained without any ESC-50 target audio.
Listening questionnaire	Three scenes with three processing conditions in each scene	Comparison between the unprocessed mixture, Base LASS, and SmartHearing.

The six sound categories in the chunk-size experiment are not intended to cover every possible danger in a real environment. They are selected as a controlled set for testing the influence of chunk size. In the complete system, the LALM uses open-language descriptions and assigns each detected sound a role, including urgent warning, other safety sound, direct speech,

user-desired content, environmental context, and unwanted noise. The system is therefore not limited to the six labels used in this experiment.

Dataset Preparation

Audio Datasets

Several public audio datasets are combined in the experiments. FSD50K provides a large collection of labelled sound events [5], while ESC-50 provides another source of environmental sounds [13]. UrbanSound8K, NIGENS, and TUT Rare Sound Events 2017 are used to include more target recordings and interference sounds.

Table 6.2: Use of the public audio datasets.

Dataset	Use in this research
FSD50K development set	Main source of the Adapter training and validation audio. Its metadata are also used to find possible target and interference sounds.
FSD50K evaluation set	Used to construct the evaluation data and not included in Adapter training.
ESC-50	Included in the multisource audio collection. It is completely removed from both training and validation when training the ESC-50-excluded Adapter.
UrbanSound8K	Provides additional urban safety sounds and environmental interference.
NIGENS	Provides environmental sounds and rare events recorded under different conditions.
TUT Rare Sound Events 2017	Provides additional transient safety-related sounds.

The initial split used dataset-specific source keys. A later cross-catalog audit found 98 benchmark mixtures sharing a Freesound recording origin with training or validation, and 38 identifiers shared between training and validation. Thus complete recording-level isolation is not established. Dataset names and different crops do not guarantee independent recordings; NIGENS/TUT origins and pretraining exposure are not fully covered by this audit. MAESTRO is not present in the actual V2 training/validation or main benchmark manifests and is not counted as an experimental source here.

Target and Interference Sounds

Adapter V2 uses seven target categories: Alarm, Siren, Vehicle Horn, Crying and Sobbing, Screaming, Glass Breaking, and Dog Barking.

The chunk-size experiment uses six of these categories: Alarm, Bark, Crying and Sobbing,

Glass Breaking, Screaming, and Siren. It also includes ten interference types: Crowd, Wind, Music, Engine, Speech, Rain, Fire, Footsteps, Knock, and Machinery.

Using several interference types prevents the result from depending on only one noise condition. Speech is included as an interference condition because human voices are not always the target sound. However, this does not mean that the system always treats speech as unwanted noise. If the LALM determines that someone is directly speaking to the user, the speech can instead be preserved or enhanced.

Table 6.3: Examples of the queries used to describe each target sound.

Target sound	Example queries
Alarm	alarm; warning alarm; alert tone
Siren	siren; emergency siren; police or ambulance siren
Vehicle Horn	car horn; vehicle horn; honking
Crying and Sobbing	person crying; baby crying; crying and sobbing
Screaming	scream; human scream; person screaming
Glass Breaking	glass breaking; glass shattering; broken glass
Dog Barking	dog barking; dog bark; barking dog

These expressions are not treated as different sound classes. They are different ways of describing the same target sound. These are descriptive query examples, not sampled training variants. The training manifests use seven fixed queries: alarm, siren, car horn, crying, scream, glass breaking, and dog barking.

Adapter Training

Adapter V1 and Adapter V2

Adapter V1 is the initial version of the experiment. It is trained mainly with FSD50K development audio and uses 1,000 training mixtures and 200 validation mixtures. Some target categories do not contain enough clean examples in this dataset. As a result, the target sounds and interference conditions used for V1 are less diverse.

Adapter V2 uses 3,500 training mixtures and 700 validation mixtures. The dataset covers seven target categories, and each category contains 500 training mixtures and 100 validation mixtures. The audio is collected from several datasets, the interference conditions are more balanced, but the later recording-origin audit found training/validation overlap. The network structure is the same in V1 and V2. Their historical comparison changes data coverage and correction-strength settings together; it does not isolate the benefit of source-level separation.

Query-space Adapter

The CLAP text encoder [4] and Base LASS [11] remain frozen when the Adapter is trained. Given the original query embedding e_k , the Adapter adds a learned correction:

$$e'_k = e_k + \alpha_k A_\theta(e_k). \quad (6-1)$$

In this equation, $A_\theta(e_k)$ is the correction learned by the Adapter, and α_k controls how much of this correction is added to the original embedding. The resulting embedding e'_k is then used by LASS. The input of the Adapter is a 512-dimensional CLAP embedding, and its bottleneck dimension is 128. Only 131,712 parameters are trained.

The final layer of the Adapter is initialized to zero. At the beginning of training, the corrected embedding is therefore the same as the original embedding. The Adapter then gradually learns the required correction.

Correction-strength recommendations are fixed for each reported evaluation run; the historical runs used different recommendation files and must be reported separately. If a query category has not been validated, the system sets $\alpha_k = 0$ and uses the original Base LASS embedding instead of applying an untested correction.

Training Loss and Settings

The loss used to train the Adapter is

$$\mathcal{L} = -\text{SI-SDR} + 0.5\mathcal{L}_1 + 0.1\mathcal{L}_{\text{MRSTFT}} + 0.01\|e'_k - e_k\|_2. \quad (6-2)$$

The negative SI-SDR term encourages the separated audio to approach the clean target. $\mathcal{L}_1$ measures the waveform difference between the output and the target. $\mathcal{L}_{\text{MRSTFT}}$ compares their spectral structures at several time–frequency resolutions. The last term keeps the corrected embedding close to the original CLAP embedding, so that the Adapter does not make an excessively large change.

For the multi-resolution STFT loss, the spectrum is calculated at three FFT resolutions: 512, 1,024, and 2,048.

The Adapter is trained with AdamW. The learning rate and weight decay are both 10^{-4} , the batch size is one, and the gradient clipping value is 5. Training lasts for at most 20 epochs and stops if the validation result does not improve for five consecutive epochs. Among the saved checkpoints, the one with the highest validation SI-SDR_i is used for the final evaluation. The random seed is 2026. For the actual positive-only training run, target-to-interference SNR is sampled uniformly between -10 and 10 dB; validation uses the saved manifest SNR. The

configured $[-5, 10]$ dB range belongs to the optional additional safety branch in three-query mode and is not used in this positive-only run.

Chunk-size Experiment

Experimental Design

The experiment contains six target categories, ten interference categories, five input SNRs $\{-10, -5, 0, 5, 10\}$ dB, and two repetitions of every target–interference–SNR combination. The number of mixed-audio samples is therefore

$$6 \times 10 \times 5 \times 2 = 600. \quad (6-3)$$

Each mixture is tested with ten chunk sizes:

$$\{2, 2.5, 3, 4, 5, 6, 7, 8, 9, 10\} \text{ s.} \quad (6-4)$$

The 600 mixtures are each evaluated with all ten chunk sizes, producing 6,000 test cases. For every combination of sound category and chunk size, the same 100 mixtures are used. The performance of different chunk sizes can therefore be compared on the same audio samples.

LALM Settings

The perception path uses qwen3.5-omni-plus-realtime-2026-03-15. Its audio input is sampled at 16 kHz, while the LASS execution path processes audio at 32 kHz. The structured prompt used in the experiment is saved as `compact_three_query_v1`.

The prompt asks the LALM to consider the complete acoustic scene. The model returns a structured description containing the detected sound, its role, danger level, confidence, and relevance to the user. Speech can therefore be preserved when it represents direct communication, while a non-speech safety sound can be given higher priority when it indicates danger.

Evaluation Metrics and Statistical Tests

SNR_i and SI-SDR_i measure how much the separated output improves over the original mixture. PESQ and STOI are also reported as additional measures of perceptual quality and intelligibility. Since PESQ and STOI were mainly developed for speech, they are treated as secondary metrics when evaluating environmental sounds. If PESQ cannot be calculated because no valid utterance is detected, the result is marked as invalid rather than being recorded as zero.

The systems are compared on the same mixtures, so the main comparisons are paired.

Confidence intervals are calculated with source-clustered bootstrap sampling. In this calculation, mixtures created from the same original recording are kept in the same group instead of being treated as completely independent samples. The number of valid samples is reported separately for each metric. The external comparison and historical ablation each contain 600 pairs (547 valid PESQ pairs) but use different input/strength settings. External means average within-target-source means, with 5,000 source-bootstrap resamples. The later LALM diagnostic uses 550 SNRi/SI-SDRi pairs and 527 PESQ/STOI pairs. Its response-time panel uses a distinct 545-request serial subset. The deep-denoising figure uses 550 environmental mixtures and 100 speech mixtures from 20 target sources. These sample sets and statistics are not pooled.

Listening Test

Listening Audio

The listening questionnaire contains three scenes. Each scene includes the unprocessed mixture, the output of Base LASS, and the output of SmartHearing. The names of the three conditions are hidden from the respondents and replaced by A, B, and C.

Table 6.4: Scenes and condition assignments in the listening questionnaire.

Scene	Target	Interference	A	B	C
1	Siren	Engine	Base LASS	Mixture	SmartHearing
2	Alarm	Rain	SmartHearing	Mixture	Base LASS
3	Crying and Sobbing	Crowd	SmartHearing	Mixture	Base LASS

Before rating each scene, the respondents first listen to a short target-only preview, which helps them understand which sound they should pay attention to. The clips used for rating are adjusted to approximately -20 dBFS RMS, with their peaks limited to -1 dBFS. The target previews are adjusted to approximately -23 dBFS RMS with a -2 dBFS peak limit. A 25-ms fade is added at both ends of each clip to avoid sudden clicks. The scored v7 SmartHearing condition uses an Adapter output with fixed target/residual gains of $+10/0$ dB and a passthrough share of 0.03 before level matching. It is a listening-specific enhancement policy, not the unmodified default system. The questionnaire statistics refer only to the v7 compact export.

Questionnaire

The compact v7 questionnaire records consent and whether headphones and a quiet environment are available. It does not record a verified hearing assessment or a separate playback-device response. In each of the three scenes, they first listen to the target preview and then listen

to the three clips labelled A, B, and C.

Each clip is rated on a seven-point scale. When giving the score, the respondents consider whether the target sound is easy to notice, whether the background remains natural, whether the audio is comfortable to listen to, and whether it contains audible artifacts. They also select the condition in which the target is easiest to notice and the condition they would prefer to use. After all three scenes, they report whether any clip caused discomfort and give an overall helpfulness score. The compact export contains no separate background-naturalness choice or optional-comment field. Respondents are instructed to stop if uncomfortable.

Results

The questionnaire contains 101 complete response records and 303 scene-level choices. SmartHearing is selected as the condition in which the target sound is easiest to notice in 50.5% of the choices. The corresponding percentages are 28.1% for Base LASS and 21.5% for the unprocessed mixture. SmartHearing is selected as the final preference in 49.5% of the choices, compared with 27.1% for Base LASS and 23.4% for the mixture.

The three conditions produced significantly different selection results for both questions. For target detectability, the Friedman test gave $p = 4.805 \times 10^{-8}$ and Kendall's $W = 0.167$. For final preference, it gave $p = 2.09 \times 10^{-5}$ and Kendall's $W = 0.107$. After Holm correction, the pairwise comparisons show that SmartHearing is selected more often than both Base LASS and the mixture for these two questions. The difference between Base LASS and the mixture is not significant.

Limitations

The local sound-separation experiments, mixture generation, metric calculation, Adapter training, and statistical analysis can be reproduced using the saved scripts, manifests, configuration files, and model checkpoints. Reproducing the LALM experiment also requires access to the same DashScope model version. Since the hosted model and its API may change, the exact model name and the saved structured outputs should be kept with the submitted code.

The experiments show controlled sound separation, dataset-held-out testing, exploratory chunk-size selection, and preliminary questionnaire records. The 101-record export contains 100 submissions within a 169-second window; recruitment and listening compliance require independent verification before claiming 101 independently validated listeners. This pattern is a verification concern, not by itself proof of invalid responses. The window policy also requires selection and evaluation on separate recordings, and complete event-to-playback response remains unmeasured. However, the current system has not been tested as a clinical hearing aid

and has not yet been implemented as a complete wearable device. Its main limitations are incorrect target selection by the LALM, the cloud-dependent latency of the perception path, the small number of scenes in the listening questionnaire, and the lack of a controlled evaluation involving verified hearing-impaired participants.